End-to-End Workflow

Based on LR-FHSS Framework and 3GPP TR 38.811

Cross-Layer Pipeline

- Orbit -> satellite position
- Coverage -> user distribution
- Load -> active devices
- Channel -> SNR / SINR
- Decoding -> successful packets
- Power -> energy consumption

Project Repository

https://github.com/SatNaingTun/LRFHSS_MultiBeam_Integration.git

Problem Statement

What Is Estimated at Each Step

Given satellite state at step t, estimate:

1. covered population $P_{\mathrm{cov}}(t)$
2. calculated nodes $N_{\mathrm{node}}(t)$
3. calculated demodulators $N_{\mathrm{demod}}(t)$
4. elevation-conditioned load and decoding behavior
5. busy/idle/sleep demod split and power

Workflow

- Set input data, simulation steps, and elevation scenarios.
- Move the satellite step by step along the orbit.
- For each step, estimate covered population.
- Convert covered population into active nodes and available demodulators.
- For each elevation case, split demodulators into busy, idle, and sleep states.
- Run LR-FHSS decoding simulation for each elevation case.
- Save output files for each step.
- Merge all steps into time-series results.
- Plot time vs decoded payloads.
- Plot time vs energy.
- Plot time vs demodulator states.

Orbit Mean Motion

$$n = \sqrt{\frac{\mu}{a^3}}$$

- Symbols: n mean motion, μ Earth gravitational parameter, a semi-major axis.
- Ref: Orbital mechanics (Vallado) https://doi.org/10.1007/978-1-4939-0802-8

Mean Anomaly Update

$$M(t) = M_0 + n(t - t_0)$$

- Symbols: $M(t)$ anomaly at time t, M_0 anomaly at reference epoch t_0, n mean motion.
- Ref: Orbital mechanics (Vallado) https://doi.org/10.1007/978-1-4939-0802-8

Horizon Central Angle

$$\psi_h = \arccos\left(\frac{R_E}{r_{\mathrm{orb}}}\right)$$

- Symbols: ψ_h horizon central angle, R_E Earth radius, r_{orb} orbital radius.
- Ref: NTN geometry context https://www.3gpp.org/DynaReport/38.811.htm

Geometric Footprint Radius

$$R_{\text{geo}} = R_E \psi_h$$

- Symbols: R_{geo} geometric footprint radius, R_E Earth radius, ψ_h horizon central angle.
- Ref: NTN geometry context https://www.3gpp.org/DynaReport/38.811.htm

Node Mapping

$$N_{\text{node}}(t) = P_{\text{pop}}\rho_{\text{node}}$$

- Symbols: $N_{\text{node}}(t)$ calculated nodes, P_{pop} population base, ρ_{node} node/population ratio.
- Ref: Implementation rule in `modules/satellite_stepper.py`

Demod Mapping

$$N_{\text{demod}}(t) = P_{\text{pop}} \rho_{\text{demod}}$$

- Config values (current defaults): $\rho_{\text{node}} = 10^{-7}$, $\rho_{\text{demod}} = 10^{-7}$.
- Symbols: $N_{\text{demod}}(t)$ calculated demodulators, P_{pop} population base, ρ_{demod} demod/population ratio.
- Ref: Implementation rule in `modules/satellite_stepper.py`

Demodulator Allocator Logic

`RecursiveReuseDemodAllocator` (**`modules/demodulator_allocator.py`**)

State model per demodulator:

- `IDLE`: no reservation.
- `BOOKED`: reserved for upcoming payload.
- `BUSY`: currently decoding payload.
- `SLEEP`: entered after idle streak exceeds threshold.

Allocation flow per frame request:

- FIFO-RR1: prefer `IDLE` demods, then `BOOKED` demods with safe gap before next start.

Elevation Node Function

Purpose:

- For each elevation in `elev_list` (e.g., 90, 79, 55, 25), estimate elevation-conditioned user geometry.

How it is used:

- `get_current_nodes_for_elevation(elev)` returns `num_nodes = elev_<token>_num_users`.
- Demod allocator flow then uses `num_users` and `distance_km` to derive busy/idle/sleep behavior.

Why Each Elevation Shows 79 Users

Current implementation in `SatelliteStepper` splits users equally across elevation bins.

Given:

- total users at current step: $N_{\text{user}} = 237$
- number of elevation bins: $N_{\text{bins}} = 3$ (90, 55, 25)

Then:

$$\text{base} = \left\lfloor \frac{237}{3} \right\rfloor = 79, \quad \text{remainder} = 0$$

So:

- `elev90_num_users = 79`, `elev55_num_users = 79`, `elev25_num_users = 79`

Why Processing Count Changes (1000 -> 748)

In LR-FHSS simulator, `node_points` means number of sampled points before integer deduplication.

Process:

1. Build logspace from `node_min` to `node_max` (default effectively 10 to 10000).
2. Convert each sampled value to `int(round(v))`.
3. Keep unique positive integers only.

Result examples from current runs:

- `node_points=1000` -> `748` unique nodes
- `node_points=10000` -> `4236` unique nodes

Meaning:

Elevation-Dependent Path Loss Model

Core Equation

In LEO links, path loss varies with elevation angle ϵ:

$$L(\epsilon) = L_{\mathrm{FSPL}}(d(\epsilon)) + L_{\mathrm{atm}}(\epsilon)$$

- ϵ: elevation angle.
- $d(\epsilon)$: elevation-dependent slant distance.
- L_{FSPL}: free-space path loss.
- L_{atm}: atmospheric attenuation.
- Ref: paper section "Elevation-Dependent Path Loss Model".

Elevation-Dependent Path Loss Model

FSPL and Atmospheric Terms

$$L_{\mathrm{FSPL}}(d) = 20\log_{10}\left(\frac{4\pi d f_c}{c}\right)$$

Atmospheric loss increases at lower elevation due to longer air path:

$$L_{\mathrm{atm}}(\epsilon) \propto \frac{1}{\sin(\epsilon)}$$

- High elevation ($\epsilon \approx 90^\circ$): shortest path, lower total loss.
- Low elevation ($\epsilon < 30^\circ$): longer path, stronger attenuation.
- Ref: Friis transmission basis https://doi.org/10.1109/JRPROC.1946.234568

Elevation Loss to System Impact

- Higher $L(\epsilon)$ reduces received signal power.
- Lower signal power decreases SNR/SINR.
- Lower SNR/SINR increases decoding difficulty and demodulator busy time.
- Longer busy time raises receiver-side energy consumption.

Power Model

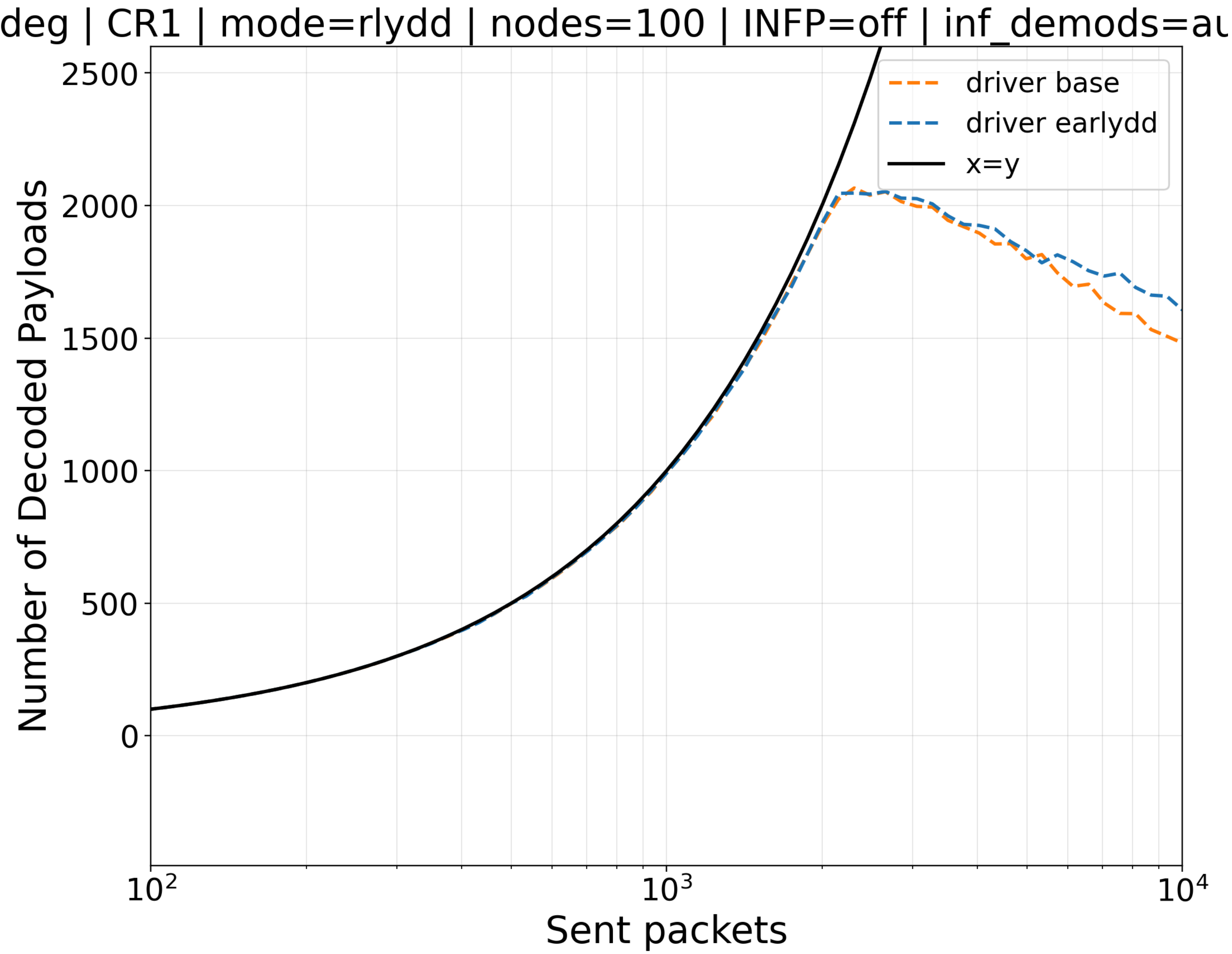
$$P_e(t) = P_0 + N_{\text{idle},e}(t)P_{\text{idle}} + N_{\text{busy},e}(t)P_{\text{busy}}$$

- Symbols: $P_e(t)$ total modeled power for elevation scenario e.
- Symbols: P_0 e-circuit power, P_{idle} idle demod power, P_{busy} busy demod power.
- Symbols: $N_{\text{idle},e}(t)$ and $N_{\text{busy},e}(t)$ are stepwise demod state counts.
- Current constants: $P_0 = 2$ mW, $P_{\text{idle}} = 9$ mW, $P_{\text{busy}} = 100$ mW.
- Ref: power-state modeling basis https://tnm.engin.umich.edu/wp-content/uploads/sites/353/2017/12/2006.10.Reducing-idle-mode-power-in-software-defined-radio-terminals_ISLPED-2006.pdf

Sparse Decode Summary Table

Busy, Idle and Energy Consumption for 100 Nodes

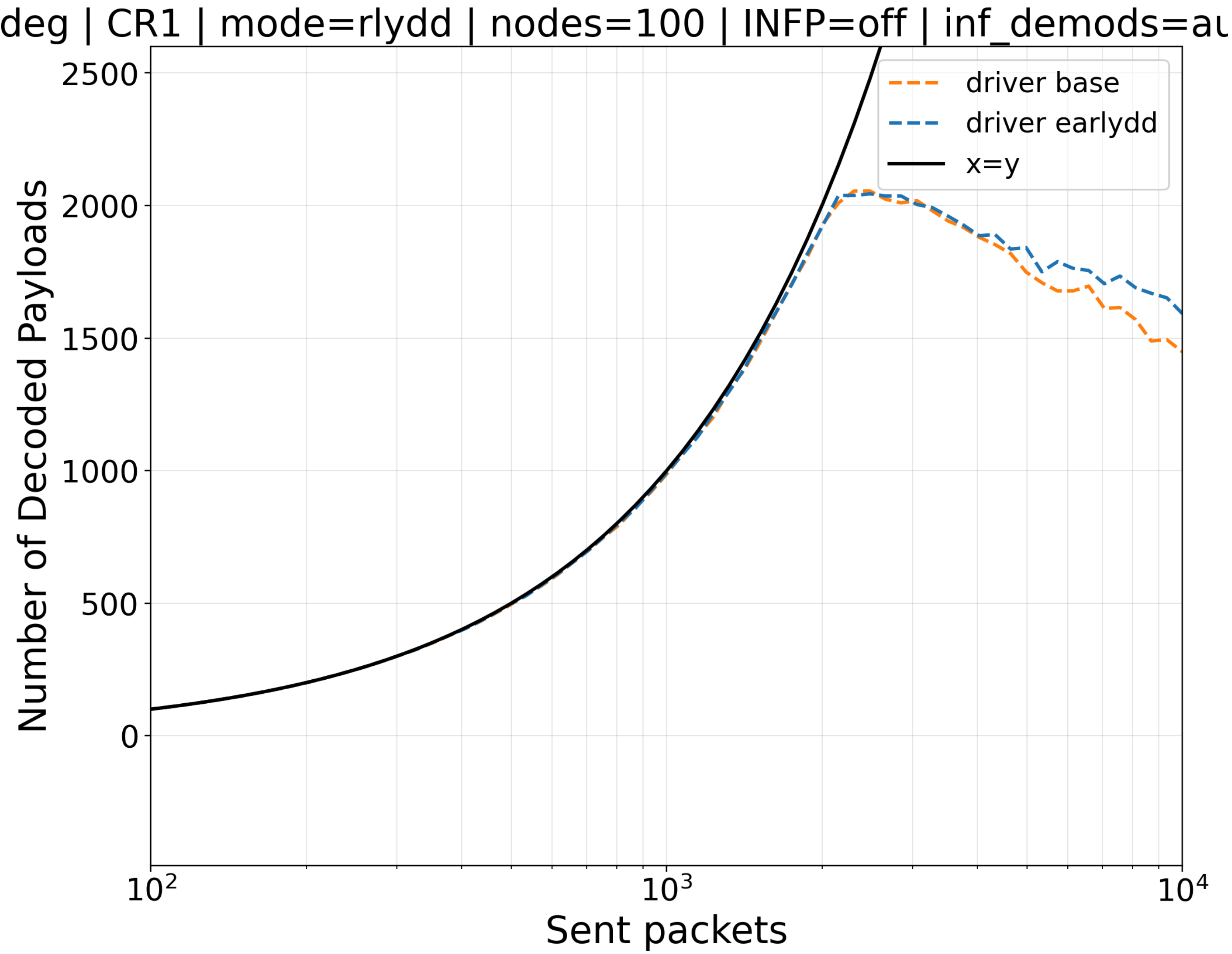
Elev.	Busy demods	Idle demods	Baseline (W)	Idle/demod (W)	Busy/demod (W)	Energy (W)
90	3	234	0.002	0.009	0.100	2.408
55	7	230	0.002	0.009	0.100	2.772
25	14	223	0.002	0.009	0.100	3.409



One-Position Sparse Decode Results

Elevation Curves (90 deg)

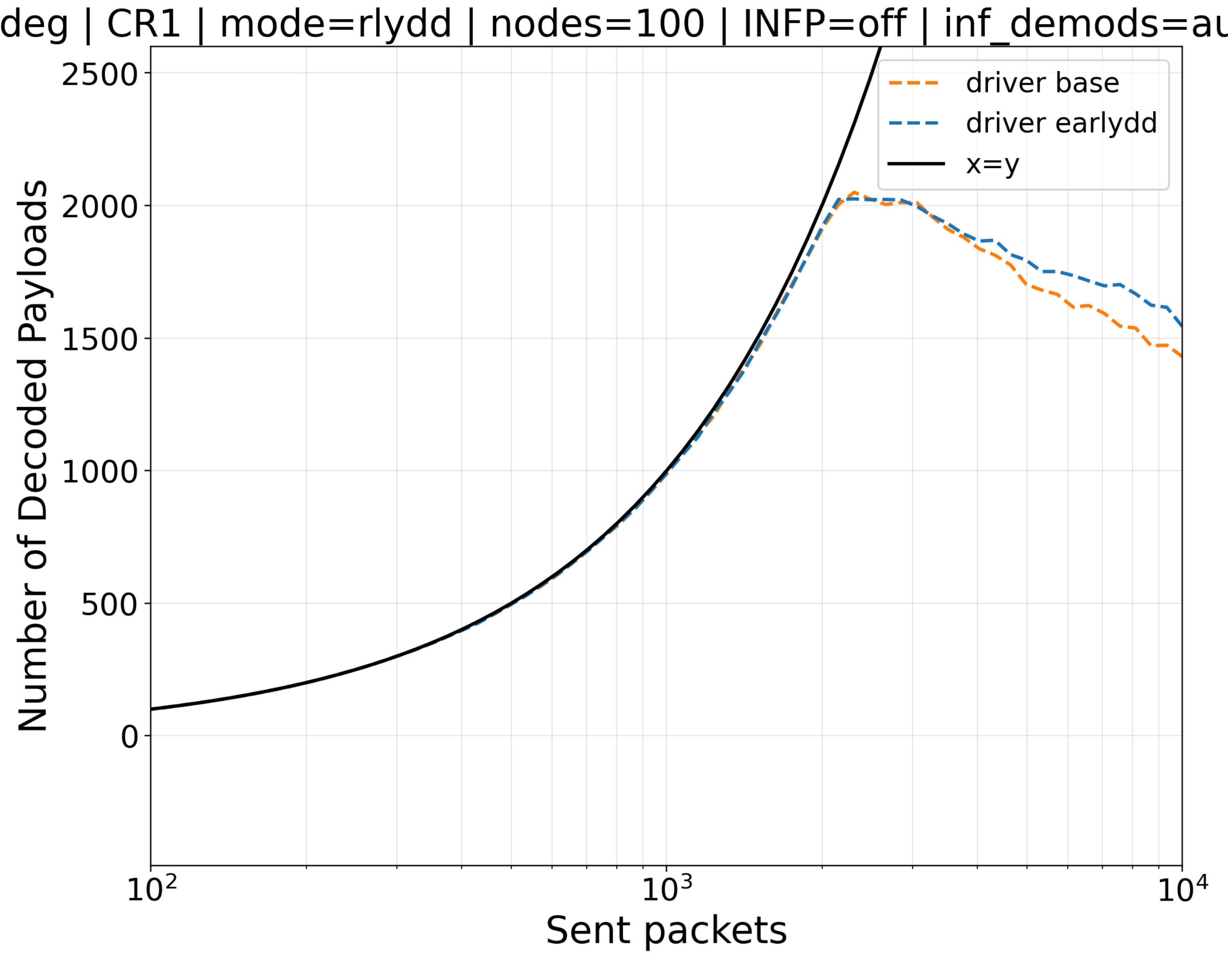
- Footprint area: 30,690.2 km^2
- Elevation: 90 deg
- Distance: 604.4 km
- Nodes 100
- Do not consider demod allocation



One-Position Sparse Decode Results

Elevation Curves (55 deg)

- Footprint area: 30,690.2 km^2
- Elevation: 55 deg
- Distance: 722.4 km
- Nodes 100
- Do not consider demod allocation



One-Position Sparse Decode Results

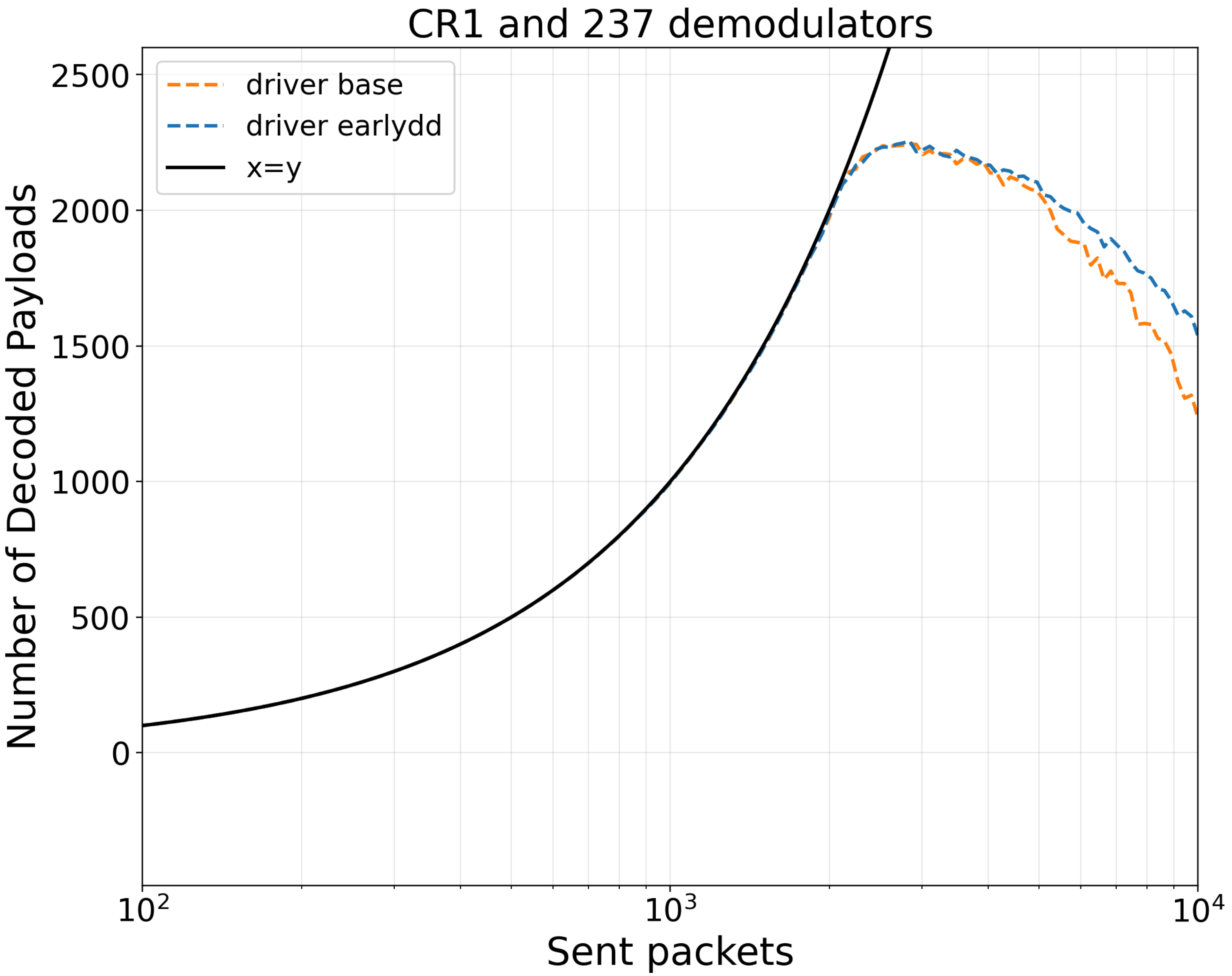
Elevation Curve (25 deg)

- Footprint area: 30,690.2 km^2
- Elevation: 25 deg
- Distance: 1203.5 km
- Nodes 100
- Do not consider demod allocation

One-Position Dense Decode Summary Table

Busy, Idle and Energy Consumption from One-Position LR-FHSS

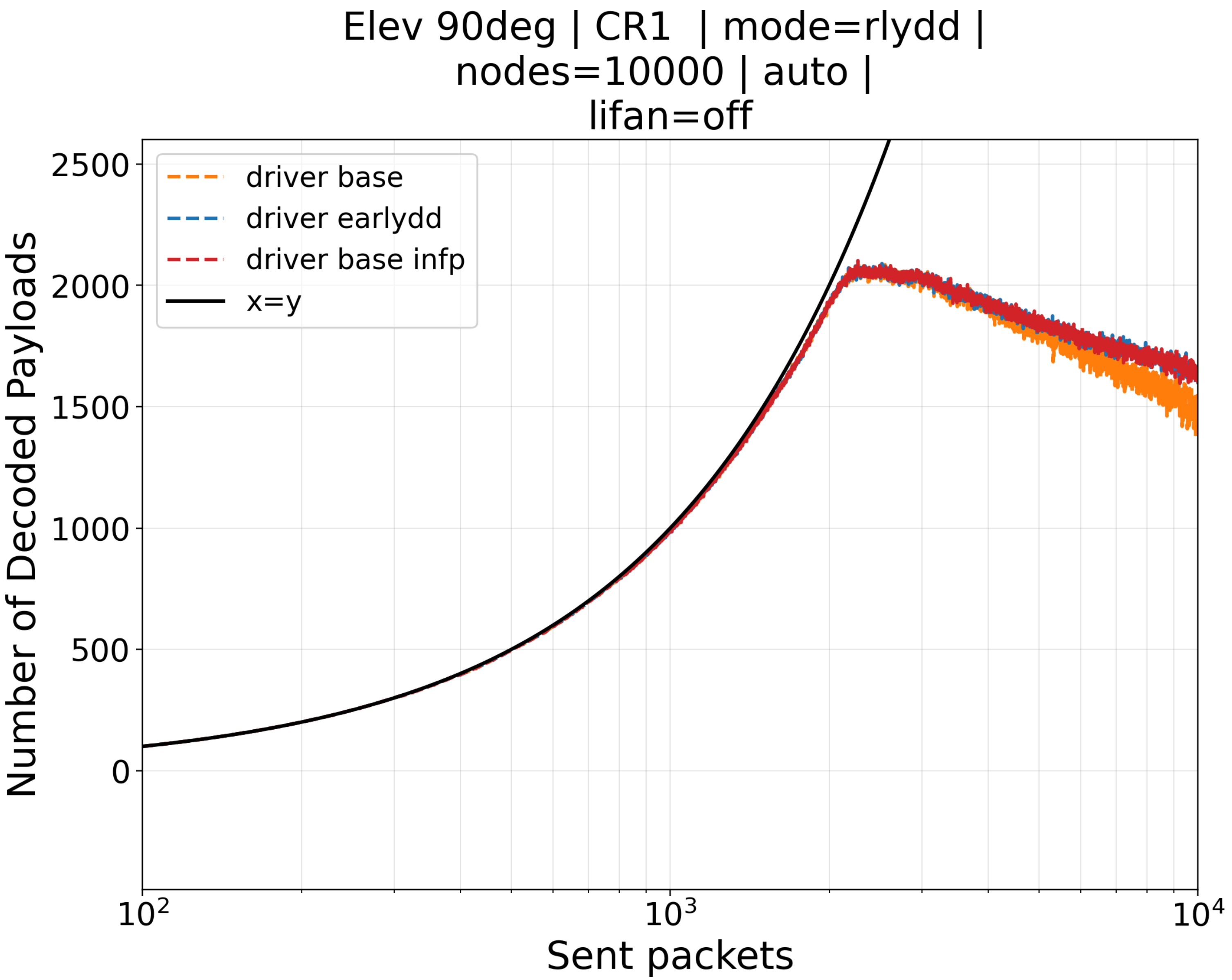
Elev.	Busy demods	Idle demods	Baseline (W)	Idle/demod (W)	Busy/demod (W)	Energy (W)
90	8	229	0.002	0.009	0.100	2.863
55	18	219	0.002	0.009	0.100	3.773
25	34	203	0.002	0.009	0.100	5.229



One-Position Dense Decode Results

Not Fixed Elevation

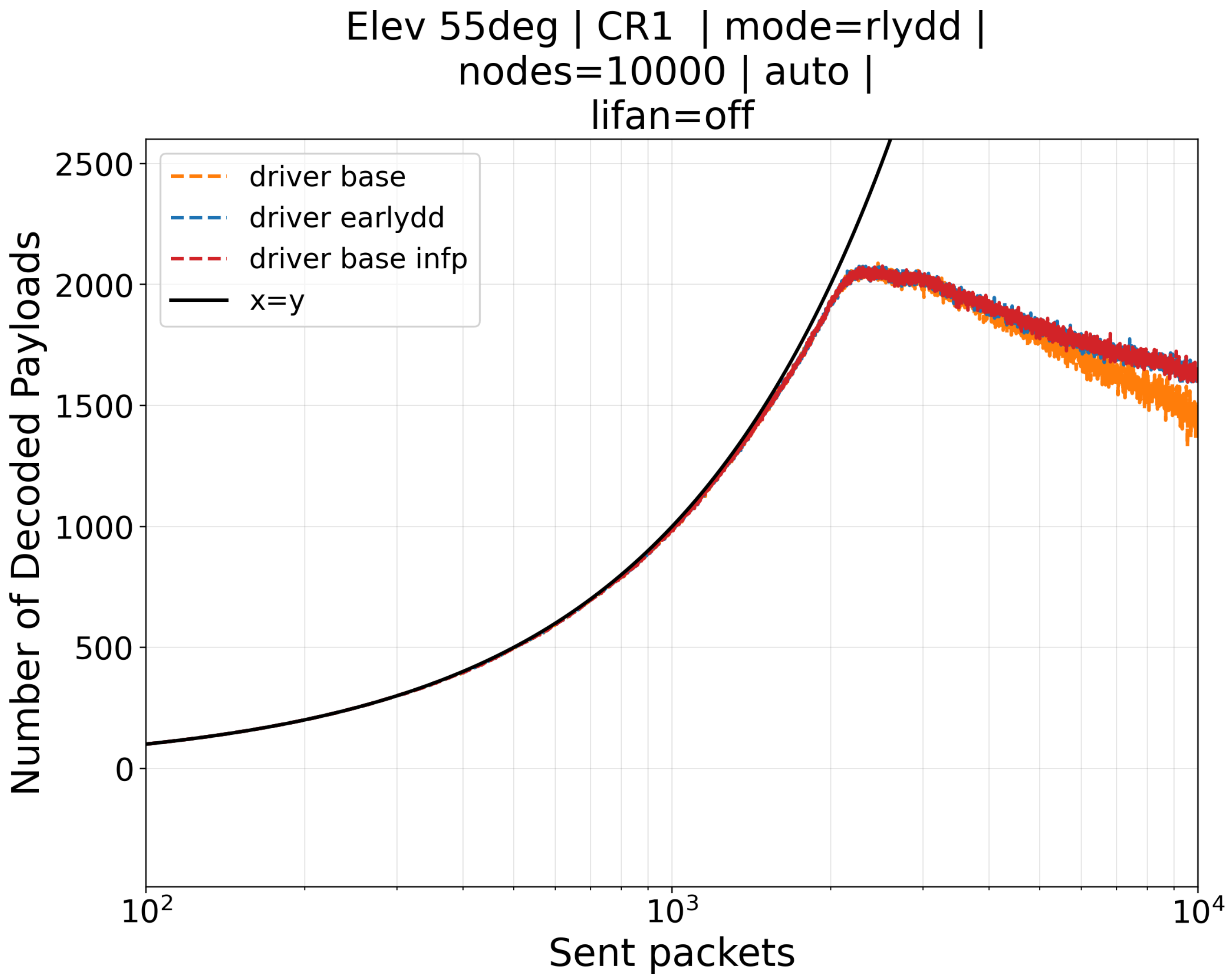
- Footprint radius: 98.8 km
- Footprint area: 30,690.2 km^2
- Mean user distance: 597.5 km
- Do not consider demod allocation



One-Position Dense Decode Results

Elevation Curves (90 deg)

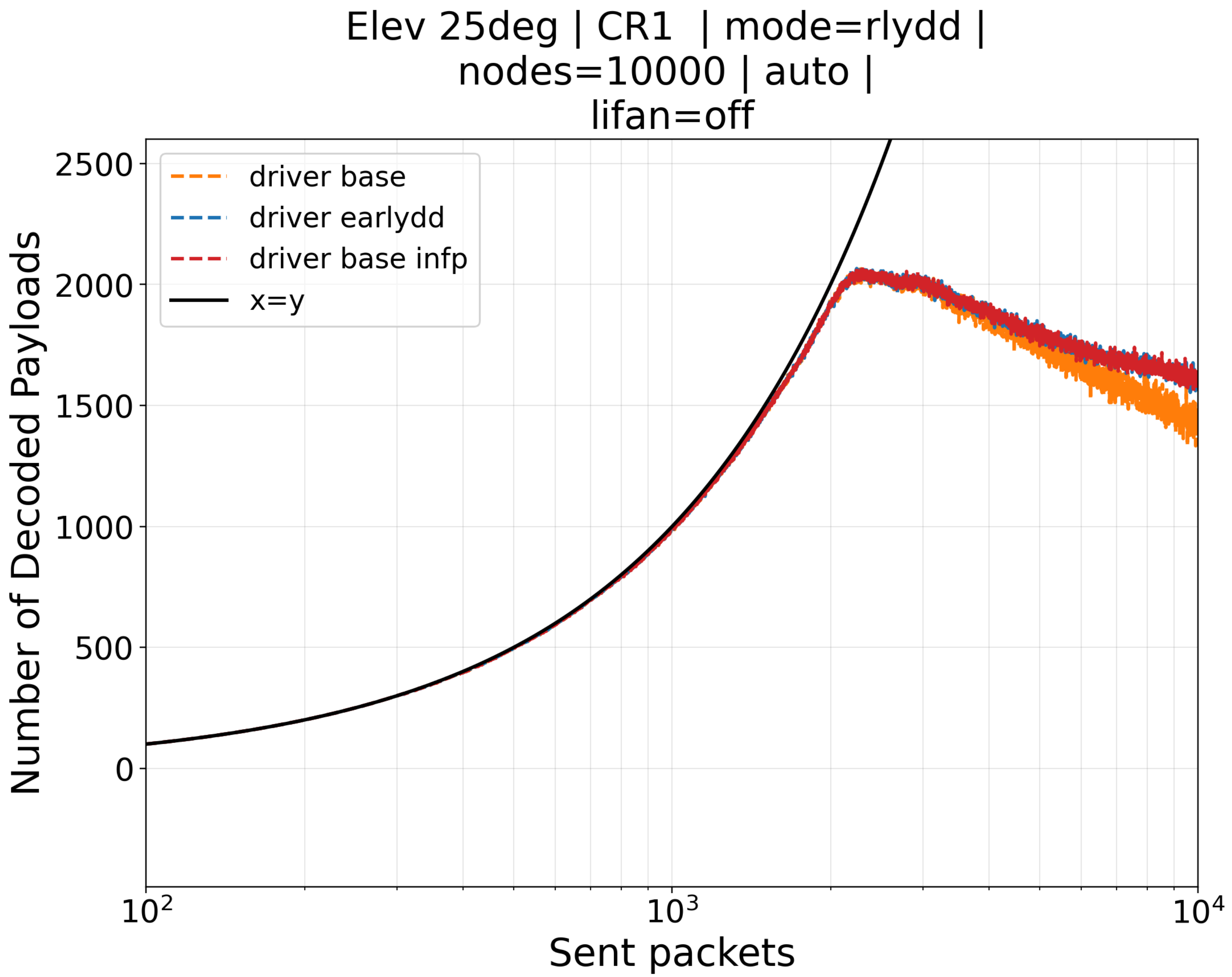
- Footprint area: 30,690.2 km^2
- Elevation: 90 deg
- Distance: 604.4 km
- Nodes 79
- Do not consider demod allocation



One-Position Dense Decode Results

Elevation Curves (55 deg)

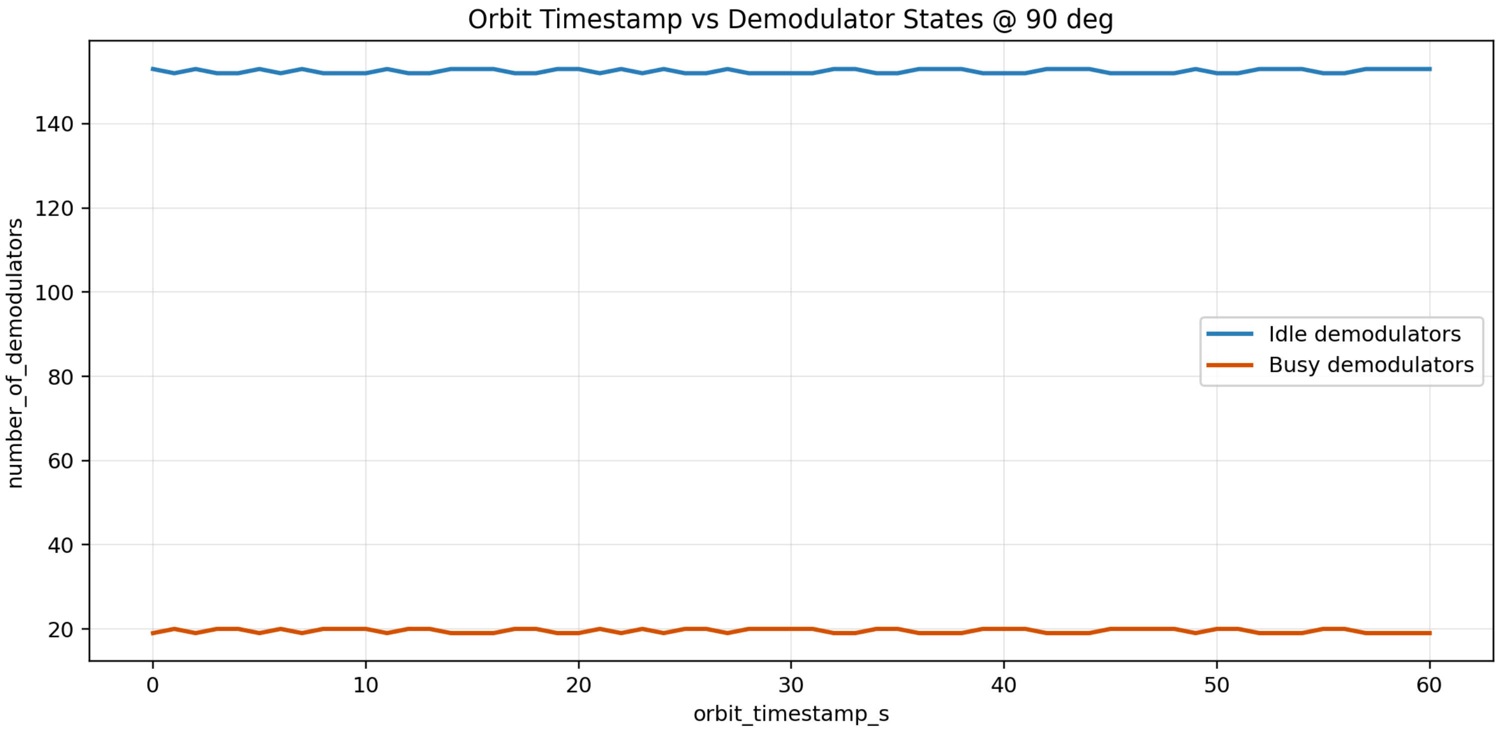
- Footprint area: 30,690.2 km^2
- Elevation: 55 deg
- Distance: 722.4 km
- Nodes 79
- Do not consider demod allocation



One-Position Dense Decode Results

Elevation Curve (25 deg)

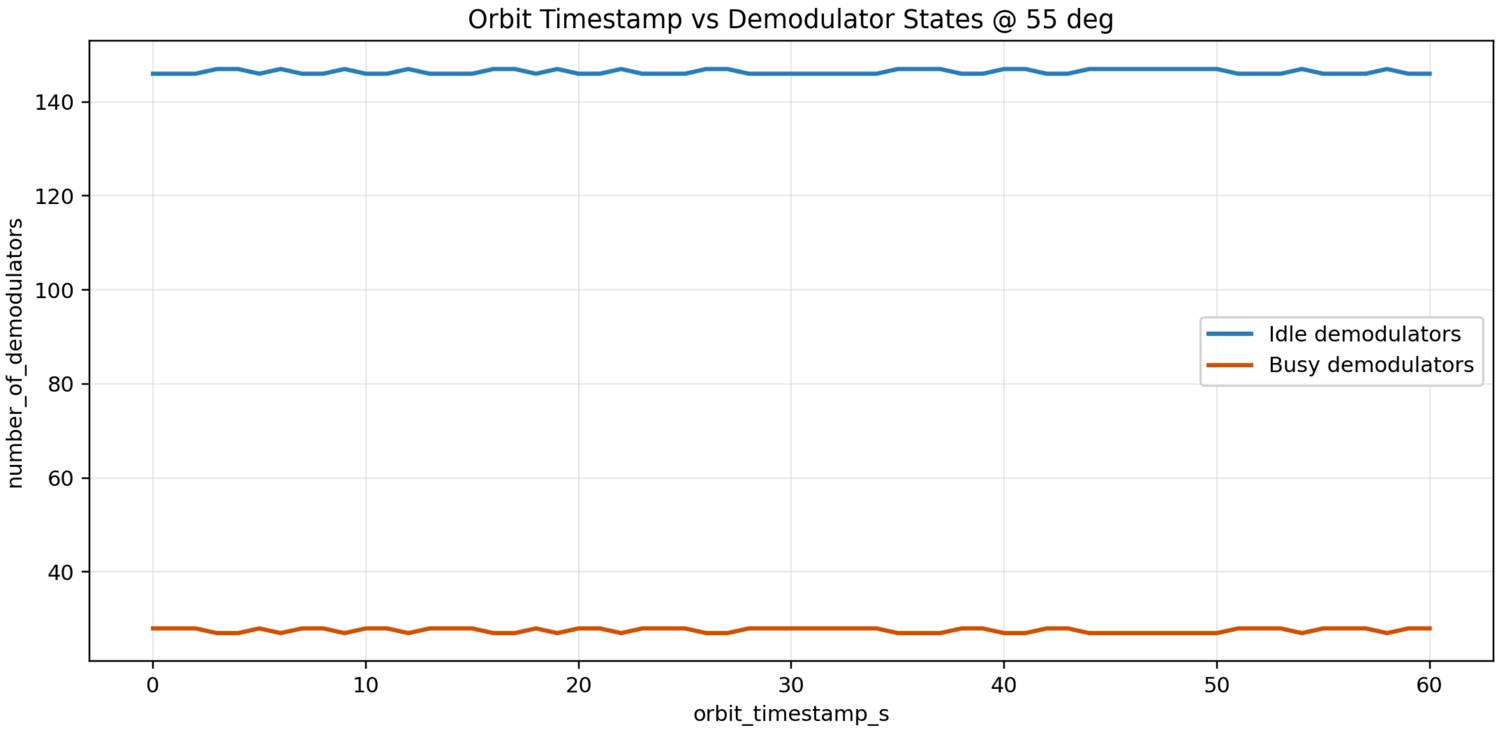
- Footprint area: 30,690.2 km^2
- Elevation: 25 deg
- Distance: 1203.5 km
- Nodes 79
- Do not consider demod allocation



Demodulator State Results

Busy/Idle vs Orbit Timestamp (90 deg)

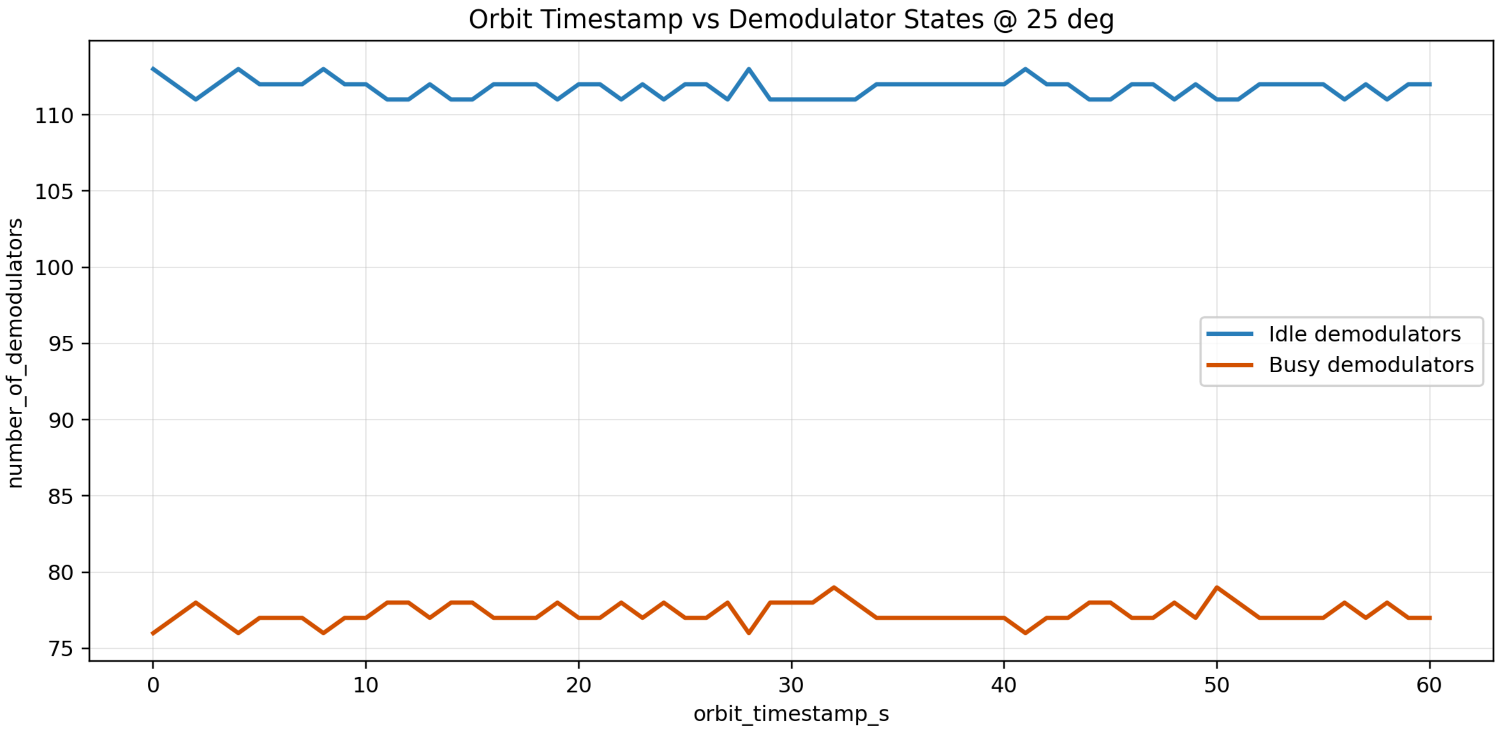
- Steps: 0-60
- Elevation: 90 deg
- Busy demods: 19-20
- Idle demods: 152-153



Demodulator State Results

Busy/Idle vs Orbit Timestamp (55 deg)

- Steps: 0-60
- Elevation: 55 deg
- Busy demods: 27-28
- Idle demods: 146-147



Demodulator State Results

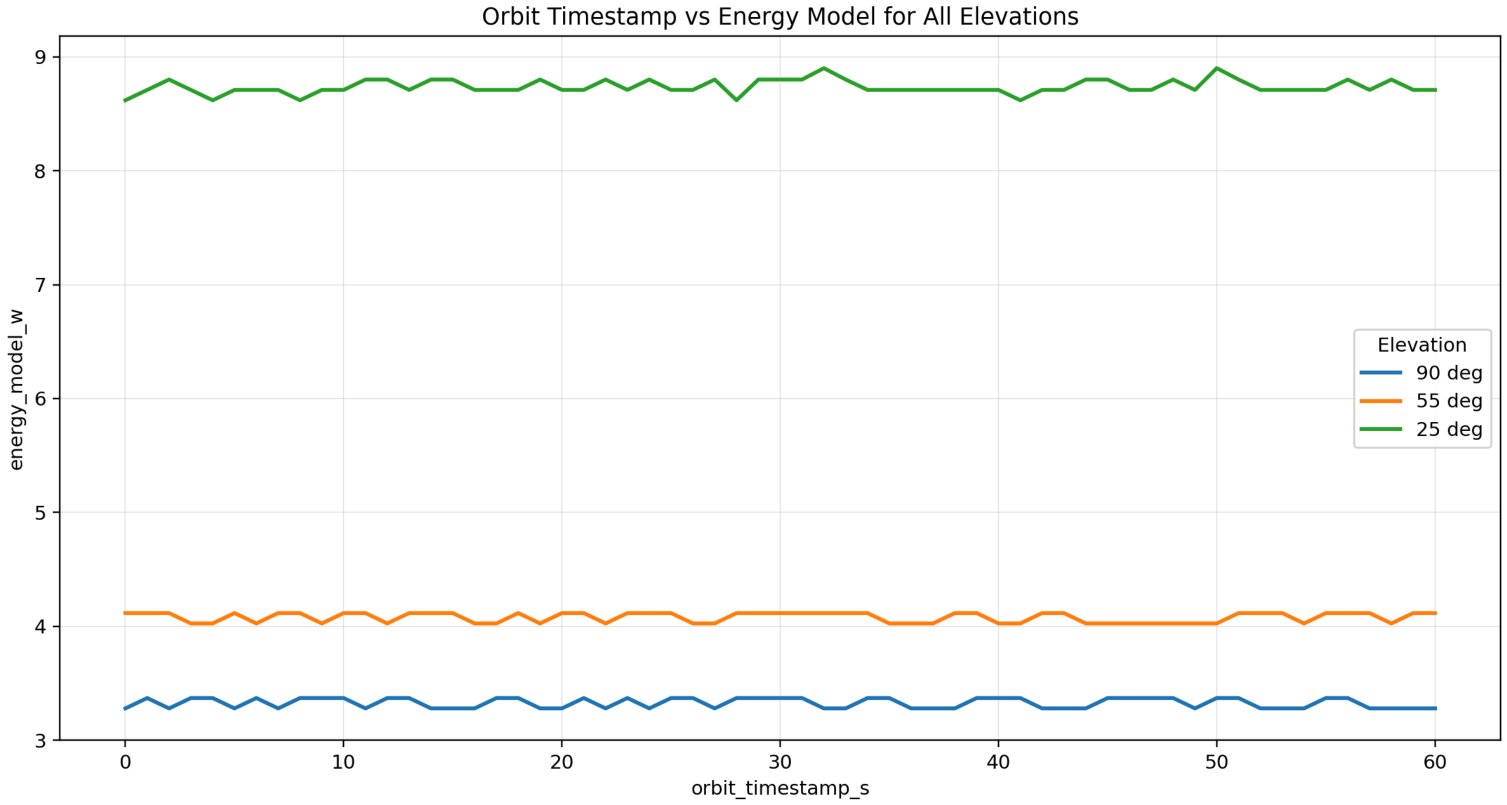
Busy/Idle vs Orbit Timestamp (25 deg)

- Steps: 0-60
- Elevation: 25 deg
- Busy demods: 76-79
- Idle demods: 111-113

Elevation Summary Table

Busy, Idle and Energy Consumption Across Steps 0-60

Elev.	Busy demods	Idle demods	Baseline (W)	Idle/demod (W)	Busy/demod (W)	Energy (W)
90	19-20	152-153	0.002	0.009	0.100	3.279-3.370
55	27-28	146-147	0.002	0.009	0.100	4.025-4.116
25	76-79	111-113	0.002	0.009	0.100	8.619-8.901



Energy Model Results (90, 55 and 25 deg)

- Steps: 0-60
- 90 deg energy: 3.279-3.370 W
- 55 deg energy: 4.025-4.116 W
- 25 deg energy: 8.619-8.901 W

Why Busy Demodulators Change with Elevation

Core Mechanism

- Total number of demodulators is fixed.
- What changes is **how long each demodulator is occupied**.

Key Insight

$$\text{Busy Demods} \propto \lambda \times T_{\text{decode}}$$

- Arrival rate ≈ similar across elevations (same nodes)
- Decoding time varies with channel quality

Interpretation

- High elevation → fast decoding → short occupation time
- Low elevation → slow decoding → long occupation time

👉 This directly controls the number of busy demodulators

Future Work

Goal: move from scenario simulation to validated, adaptive NTN system design.

1. Validate with real data: calibrate traffic, channel loss, and demod behavior using measurements.
2. Add uncertainty modeling: include weather, bursty traffic, and mobility distributions with confidence intervals.
3. Compare access schemes: benchmark LR-FHSS against alternative PHY/MAC options under equal constraints.

Future Work

4. Optimize control: design dynamic demod/power allocation policies using optimization or reinforcement learning.
5. Scale to network level: extend from single-satellite view to multi-satellite and inter-satellite coordination.
6. Build reproducible benchmarks: publish datasets, metrics, and open evaluation protocols for fair comparison.

Citable Paper Sources (URLs)

- LR-FHSS overview paper: https://doi.org/10.1109/MCOM.001.2000627
- 3GPP NTN reference (TR 38.811): https://www.3gpp.org/DynaReport/38.811.htm
- Friis transmission formula: https://doi.org/10.1109/JRPROC.1946.234568
- Shannon communication theory: https://doi.org/10.1002/j.1538-7305.1948.tb01338.x
- ALOHA system paper: https://doi.org/10.1145/1478462.1478502

Citable Data Sources (URLs)

- Natural Earth populated places: https://naciscdn.org/naturalearth/10m/cultural/ne_10m_populated_places.zip
- Natural Earth countries: https://naciscdn.org/naturalearth/10m/cultural/ne_10m_admin_0_countries.zip
- Natural Earth lakes: https://naciscdn.org/naturalearth/10m/physical/ne_10m_lakes.zip
- Natural Earth rivers: https://naciscdn.org/naturalearth/10m/physical/ne_10m_rivers_lake_centerlines.zip
- Demod power baseline reference: https://tnm.engin.umich.edu/wp-content/uploads/sites/353/2017/12/2006.10.Reducing-idle-mode-power-in-